



Guide 2– BLE Communication

Objectives:

- Configuration of BLE parameters on the Raspberry
- Reading of sensor sensors connected to the Raspberry and communication of readings to the Raspberry.

1. 1. Change the hostname of the Bluetooth client (so that each PC of each colleague has a different name):

```
sudo nano /etc/hostname
```

2. Instalations of the necessary tools

```
sudo apt-get install bluez  
sudo apt install bluez-hcidump
```

3. Preparation of the blues to allow the use of BLE

```
#configure experimental bluetooth features  
sudo nano /lib/systemd/system/bluetooth.service  
#change ExecStart to  
ExecStart=/usr/lib/bluetooth/bluetoothd --experimental
```

4. Alteração do modo de funcionamento do cliente Bluetooth para a utilização de modo *Low Energy*

```
sudo nano /etc/bluetooth/main.conf  
#change line  
ControllerMode = le
```

5. Procure as interfaces BLE existentes na sua máquina

```
root@cinnamon:/home/pasg# hcitool dev  
Devices:  
hci0 34:E6:AD:42:43:74
```

6. Faça uma procura de dispositivos Low Energy

```
root@cinnamon:/home/pasg# hcitool lescan  
LE Scan ...  
D8:E0:E1:11:61:99 (unknown)  
56:6F:CA:22:F6:3B (unknown)  
E7:E5:B4:A0:72:BE (unknown)  
0E:D2:0E:57:69:9A (unknown)  
2C:B8:81:F7:BD:BA (unknown)  
E6:E4:B3:9F:71:BD (unknown)
```



```
D8:E0:E1:11:61:99 [TV] Samsung 7 Series (65)
BA:03:64:29:12:90 (unknown)
FD:6A:05:76:20:3F (unknown)
DA:1D:9A:0C:DC:D2 (unknown)
FD:6A:05:76:20:3F Mi Band 3
C0:61:34:59:18:04 G20_1804
18:BF:2F:B9:2F:CD (unknown)
C0:61:34:59:18:04 (unknown)
E6:E4:B3:9F:71:BD (unknown)
DD:41:EB:C6:E0:33 (unknown)
BA:03:64:29:12:90 MSM3
D7:BA:36:EB:4B:CD (unknown)
```

7. Ligue-se ao seu dispositivo BLE utilizando o *gatttool*

```
root@cinnamon:/home/pasg# gatttool -b BA:03:64:29:12:90 -I
[BA:03:64:29:12:90][LE]> connect
Attempting to connect to BA:03:64:29:12:90
Connection successful
```

8. Peça para aceder aos UUIDs primários. O comando deverá gerar um alista de todos os serviços disponíveis no dispositivo BLE.

```
[BA:03:64:29:12:90][LE]> primary
attr handle: 0x0001, end grp handle: 0x0005 uuid: 00001800-0000-1000-8000-00805f9b34fb
attr handle: 0x000c, end grp handle: 0x000f uuid: 00001801-0000-1000-8000-00805f9b34fb
attr handle: 0x0010, end grp handle: 0x001a uuid: 00001812-0000-1000-8000-00805f9b34fb
attr handle: 0x001b, end grp handle: 0x0020 uuid: 0000fd00-0000-1000-8000-00805f9b34fb
attr handle: 0x0021, end grp handle: 0x0026 uuid: 0000ffe0-0000-1000-8000-00805f9b34fb
attr handle: 0x0027, end grp handle: 0x0034 uuid: 0000fee7-0000-1000-8000-00805f9b34fb
attr handle: 0x0035, end grp handle: 0x0042 uuid: 0000cc00-0000-1000-8000-00805f9b34fb
```

9. Peça todos os *handles* disponíveis:

```
[BA:03:64:29:12:90][LE]> char-desc
handle: 0x0001, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x0002, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0003, uuid: 00002a00-0000-1000-8000-00805f9b34fb
handle: 0x0004, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0005, uuid: 00002a01-0000-1000-8000-00805f9b34fb
handle: 0x000c, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x000d, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x000e, uuid: 00002a05-0000-1000-8000-00805f9b34fb
handle: 0x000f, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0010, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x0011, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0012, uuid: 00002a4a-0000-1000-8000-00805f9b34fb
handle: 0x0013, uuid: 00002803-0000-1000-8000-00805f9b34fb
```



```
handle: 0x0014, uuid: 00002a4c-0000-1000-8000-00805f9b34fb
handle: 0x0015, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0016, uuid: 00002a4b-0000-1000-8000-00805f9b34fb
handle: 0x0017, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0018, uuid: 00002a4d-0000-1000-8000-00805f9b34fb
handle: 0x0019, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x001a, uuid: 00002908-0000-1000-8000-00805f9b34fb
handle: 0x001b, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x001c, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x001d, uuid: 0000fd01-0000-1000-8000-00805f9b34fb
handle: 0x001e, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x001f, uuid: 0000fd02-0000-1000-8000-00805f9b34fb
handle: 0x0020, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0021, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x0022, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0023, uuid: 0000ffe1-0000-1000-8000-00805f9b34fb
handle: 0x0024, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0025, uuid: 0000ffe2-0000-1000-8000-00805f9b34fb
handle: 0x0026, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0027, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x0028, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0029, uuid: 0000fec7-0000-1000-8000-00805f9b34fb
handle: 0x002a, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x002b, uuid: 0000fec8-0000-1000-8000-00805f9b34fb
handle: 0x002c, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x002d, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x002e, uuid: 0000fec9-0000-1000-8000-00805f9b34fb
handle: 0x002f, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0030, uuid: 0000fea1-0000-1000-8000-00805f9b34fb
handle: 0x0031, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0032, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0033, uuid: 0000fea2-0000-1000-8000-00805f9b34fb
handle: 0x0034, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0035, uuid: 00002800-0000-1000-8000-00805f9b34fb
handle: 0x0036, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0037, uuid: 0000cc02-0000-1000-8000-00805f9b34fb
handle: 0x0038, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0039, uuid: 0000cc03-0000-1000-8000-00805f9b34fb
handle: 0x003a, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x003b, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x003c, uuid: 0000cc04-0000-1000-8000-00805f9b34fb
handle: 0x003d, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x003e, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x003f, uuid: 0000cc05-0000-1000-8000-00805f9b34fb
```



```
handle: 0x0040, uuid: 00002902-0000-1000-8000-00805f9b34fb
handle: 0x0041, uuid: 00002803-0000-1000-8000-00805f9b34fb
handle: 0x0042, uuid: 0000cc06-0000-1000-8000-00805f9b34fb
[BA:03:64:29:12:90] [LE]>
```

10. Operação de leitura do valor de um *handle*

```
[BA:03:64:29:12:90] [LE]> char-read-hnd 0x0035
```

Characteristic value/descriptor: 00 cc

11. O resultado é devolvido em hexadecimal, pelo que em decimal, é igual a 24. No caso, esse número é a percentagem de carga restante da bateria no dispositivo, de acordo com a definição do serviço de bateria.

```
[BA:03:64:29:12:90] [LE]> char-read-hnd 0x0001
```

Characteristic value/descriptor: 00 18

12. O funcionamento dos serviços de notificações requer que estas sejam ativadas pelo cliente, através da escrita de uma propriedade. Após a ativação o servidor realiza o envio periódico e autónomo da informação. Active a notificação do valor da bateria colocando o *handle* com o valor 0100.

```
[BA:03:64:29:12:90] [LE]> char-write-req 0x000f 0100
```

Characteristic value was written successfully

Notification handle = 0x0039 value: 32 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 3c 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 0a 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 14 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 1e 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 28 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

```
[BA:03:64:29:12:90] [LE]>
```

13. O processo contrário, desligar o serviço de notificações, realiza-se escrevendo na propriedade o valor 0000.

Notification handle = 0x0039 value: 32 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

Notification handle = 0x0039 value: 3c 13 06 01 0c 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00

```
[BA:03:64:29:12:90] [LE]> char-write-req 0x000f 0000
```

Characteristic value was written successfully

```
[BA:03:64:29:12:90] [LE]>
```

14. Da mesma forma o gatttool permite ler todas as características do dispositivo.

```
[BA:03:64:29:12:90] [LE]> characteristics
```

handle: 0x0002, char properties: 0x02, char value handle: 0x0003, uuid: 00002a00-0000-1000-8000-00805f9b34fb



```
handle: 0x0004, char properties: 0x02, char value handle: 0x0005, uuid: 00002a01-0000-1000-8000-00805f9b34fb
handle: 0x000d, char properties: 0x20, char value handle: 0x000e, uuid: 00002a05-0000-1000-8000-00805f9b34fb
handle: 0x0011, char properties: 0x02, char value handle: 0x0012, uuid: 00002a4a-0000-1000-8000-00805f9b34fb
handle: 0x0013, char properties: 0x04, char value handle: 0x0014, uuid: 00002a4c-0000-1000-8000-00805f9b34fb
handle: 0x0015, char properties: 0x02, char value handle: 0x0016, uuid: 00002a4b-0000-1000-8000-00805f9b34fb
handle: 0x0017, char properties: 0x12, char value handle: 0x0018, uuid: 00002a4d-0000-1000-8000-00805f9b34fb
handle: 0x001c, char properties: 0x04, char value handle: 0x001d, uuid: 0000fd01-0000-1000-8000-00805f9b34fb
handle: 0x001e, char properties: 0x14, char value handle: 0x001f, uuid: 0000fd02-0000-1000-8000-00805f9b34fb
handle: 0x0022, char properties: 0x08, char value handle: 0x0023, uuid: 0000ffe1-0000-1000-8000-00805f9b34fb
handle: 0x0024, char properties: 0x12, char value handle: 0x0025, uuid: 0000ffe2-0000-1000-8000-00805f9b34fb
handle: 0x0028, char properties: 0x08, char value handle: 0x0029, uuid: 0000fec7-0000-1000-8000-00805f9b34fb
handle: 0x002a, char properties: 0x20, char value handle: 0x002b, uuid: 0000fec8-0000-1000-8000-00805f9b34fb
handle: 0x002d, char properties: 0x02, char value handle: 0x002e, uuid: 0000fec9-0000-1000-8000-00805f9b34fb
handle: 0x002f, char properties: 0x32, char value handle: 0x0030, uuid: 0000fea1-0000-1000-8000-00805f9b34fb
handle: 0x0032, char properties: 0x2a, char value handle: 0x0033, uuid: 0000fea2-0000-1000-8000-00805f9b34fb
handle: 0x0036, char properties: 0x0a, char value handle: 0x0037, uuid: 0000cc02-0000-1000-8000-00805f9b34fb
handle: 0x0038, char properties: 0x12, char value handle: 0x0039, uuid: 0000cc03-0000-1000-8000-00805f9b34fb
handle: 0x003b, char properties: 0x12, char value handle: 0x003c, uuid: 0000cc04-0000-1000-8000-00805f9b34fb
handle: 0x003e, char properties: 0x1a, char value handle: 0x003f, uuid: 0000cc05-0000-1000-8000-00805f9b34fb
handle: 0x0041, char properties: 0x08, char value handle: 0x0042, uuid: 0000cc06-0000-1000-8000-00805f9b34fb
[BA:03:64:29:12:90][LE]> quit
```

15. O bluetoothctl é uma ferramenta que permite o controlo iterativo do bluez. A lista de experiências abaixo ilustra um conjunto de funcionalidades da ferramenta.

```
root@cinnamon:/home/pasg# bluetoothctl
Agent registered
[CHG] Device 00:18:09:8B:22:48 Connected: yes
[CHG] Device 00:18:09:8B:22:48 ServicesResolved: yes
[WH-CH500]# connect BA:03:64:29:12:90
Device BA:03:64:29:12:90 not available
```

16. Realizar pesquisa dos dispositivos existentes:



```
[WH-CH500]# scan on
Discovery started
[CHG] Controller 34:E6:AD:42:43:74 Discovering: yes
[NEW] Device E6:E4:B3:9F:71:BD E6-E4-B3-9F-71-BD
[CHG] Device 78:5B:93:00:C5:A9 RSSI: -77
[CHG] Device 78:5B:93:00:C5:A9 TxPower: 12
[CHG] Device 78:5B:93:00:C5:A9 AdvertisingFlags:
    1a
[CHG] Device C0:61:34:59:18:04 RSSI: -66
[CHG] Device D8:E0:E1:11:61:99 RSSI: -80
[CHG] Device D8:E0:E1:11:61:99 AdvertisingFlags:
    1a
[CHG] Device FD:6A:05:76:20:3F RSSI: -47
[NEW] Device BA:03:64:29:12:90 MSM3
[CHG] Device D8:E0:E1:11:61:99 AdvertisingFlags:
    00
[CHG] Device D9:1C:99:0B:DB:D1 RSSI: -91
[CHG] Device E7:EB:88:EB:88:5D RSSI: -86
[NEW] Device D7:BA:36:EB:4B:CD D7-BA-36-EB-4B-CD
```

17. Parar a pesquisa de dispositivos.

```
[WH-CH500]# scan off
Discovery stopped
[CHG] Controller 34:E6:AD:42:43:74 Discovering: no
[CHG] Device D7:BA:36:EB:4B:CD RSSI is nil
[CHG] Device E7:EB:88:EB:88:5D RSSI is nil
[CHG] Device D9:1C:99:0B:DB:D1 RSSI is nil
[CHG] Device BA:03:64:29:12:90 RSSI is nil
[CHG] Device FD:6A:05:76:20:3F RSSI is nil
[CHG] Device D8:E0:E1:11:61:99 RSSI is nil
[CHG] Device C0:61:34:59:18:04 RSSI is nil
[CHG] Device 78:5B:93:00:C5:A9 TxPower is nil
[CHG] Device 78:5B:93:00:C5:A9 RSSI is nil
[CHG] Device E6:E4:B3:9F:71:BD RSSI is nil
```

18. Enumerar os dispositivos encontrados.

```
[WH-CH500]# devices
Device 88:D0:39:8C:4F:5D JBL E65BTNC
Device FD:6A:05:76:20:3F Mi Band 3
Device 00:18:56:79:25:3B THOR 4 PRO
Device DC:BB:42:22:B6:7B i30
Device C0:61:34:59:18:04 G20_1804
```



```
Device 00:18:09:8B:22:48 WH-CH500
Device 60:45:CB:2E:03:DC ASUS_Z017D
Device D8:E0:E1:11:61:99 [TV] Samsung 7 Series (65)
Device 78:5B:93:00:C5:A9 78-5B-93-00-C5-A9
Device E7:EB:88:EB:88:5D Amazfit GTS
Device D9:1C:99:0B:DB:D1 D9-1C-99-0B-DB-D1
Device E6:E4:B3:9F:71:BD E6-E4-B3-9F-71-BD
Device BA:03:64:29:12:90 MSM3
Device D7:BA:36:EB:4B:CD D7-BA-36-EB-4B-CD
```

19. Permite identificar todos os serviços do dispositivo.

```
[WH-CH500]# connect BA:03:64:29:12:90
Attempting to connect to BA:03:64:29:12:90
[CHG] Device BA:03:64:29:12:90 Connected: yes
Connection successful
[NEW] Primary Service
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service000c
    00001801-0000-1000-8000-00805f9b34fb
    Generic Attribute Profile
[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service000c/char000d
    00002a05-0000-1000-8000-00805f9b34fb
    Service Changed
[NEW] Descriptor
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service000c/char000d/desc000f
    00002902-0000-1000-8000-00805f9b34fb
    Client Characteristic Configuration
[NEW] Primary Service
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service001b
    0000fd00-0000-1000-8000-00805f9b34fb
    Unknown
[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001c
    0000fd01-0000-1000-8000-00805f9b34fb
    Unknown
[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001e
    0000fd02-0000-1000-8000-00805f9b34fb
    Unknown
[NEW] Descriptor
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001e/desc0020
    00002902-0000-1000-8000-00805f9b34fb
    Client Characteristic Configuration
```



```
[NEW] Primary Service
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0021
    0000ffe0-0000-1000-8000-00805f9b34fb
    Unknown

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0022
    0000ffel-0000-1000-8000-00805f9b34fb
    Unknown

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0024
    0000ffe2-0000-1000-8000-00805f9b34fb
    Unknown

[NEW] Descriptor
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0024/desc0026
    00002902-0000-1000-8000-00805f9b34fb
    Client Characteristic Configuration

[NEW] Primary Service
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027
    0000fee7-0000-1000-8000-00805f9b34fb
    Tencent Holdings Limited

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0028
    0000fec7-0000-1000-8000-00805f9b34fb
    Apple, Inc.

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a
    0000fec8-0000-1000-8000-00805f9b34fb
    Apple, Inc.

[NEW] Descriptor
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a/desc002c
    00002902-0000-1000-8000-00805f9b34fb
    Client Characteristic Configuration

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002d
    0000fec9-0000-1000-8000-00805f9b34fb
    Apple, Inc.

[NEW] Characteristic
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002f
    0000feal-0000-1000-8000-00805f9b34fb
    Intrepid Control Systems, Inc.

[NEW] Descriptor
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002f/desc0031
    00002902-0000-1000-8000-00805f9b34fb
    Client Characteristic Configuration
```



[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0032
0000fea2-0000-1000-8000-00805f9b34fb
Intrepid Control Systems, Inc.

[NEW] Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0032/desc0034
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

[NEW] Primary Service

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035
0000cc00-0000-1000-8000-00805f9b34fb
Unknown

[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0036
0000cc02-0000-1000-8000-00805f9b34fb
Unknown

[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0038
0000cc03-0000-1000-8000-00805f9b34fb
Unknown

[NEW] Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0038/desc003a
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003b
0000cc04-0000-1000-8000-00805f9b34fb
Unknown

[NEW] Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003b/desc003d
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003e
0000cc05-0000-1000-8000-00805f9b34fb
Unknown

[NEW] Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003e/desc0040
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

[NEW] Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0041
0000cc06-0000-1000-8000-00805f9b34fb
Unknown



20. O *menu gatt* permite ainda um conjunto estendido de funcionalidades como ler, escrever, receber notificações,

```
[MSM3]# menu gatt
```

```
Menu gatt:
```

```
Available commands:
```

```
-----
```

list-attributes [dev]	List attributes
select-attribute <attribute/UUID>	Select attribute
attribute-info [attribute/UUID]	Select attribute
read [offset]	Read attribute value
write <data=xx xx ...> [offset]	Write attribute value
acquire-write	Acquire Write file descriptor
release-write	Release Write file descriptor
acquire-notify	Acquire Notify file descriptor
release-notify	Release Notify file descriptor
notify <on/off>	Notify attribute value
register-application [UUID ...]	Register profile to connect
unregister-application	Unregister profile
register-service <UUID>	Register application service.
unregister-service <UUID/object>	Unregister application service
register-includes <UUID>	Register as Included service in.
unregister-includes <Service-UUID><Inc-UUID>	Unregister Included service.
register-characteristic <UUID> <Flags=read,write,notify...>	Register application characteristic
unregister-characteristic <UUID/object>	Unregister application characteristic
register-descriptor <UUID> <Flags=read,write...>	Register application descriptor
unregister-descriptor <UUID/object>	Unregister application descriptor
back	Return to main menu
version	Display version
quit	Quit program
exit	Quit program
help	Display help about this program
export	Print environment variables

21. Ler os atributos do dispositivo.

```
[MSM3]# list-attributes
```

```
Primary Service
```

```
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service000c
```

```
    00001801-0000-1000-8000-00805f9b34fb
```

```
    Generic Attribute Profile
```

```
Characteristic
```

```
    /org/bluez/hci0/dev_BA_03_64_29_12_90/service000c/char000d
```

```
    00002a05-0000-1000-8000-00805f9b34fb
```

```
    Service Changed
```



Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service000c/char000d/desc000f
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Primary Service

/org/bluez/hci0/dev_BA_03_64_29_12_90/service001b
0000fd00-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001c
0000fd01-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001e
0000fd02-0000-1000-8000-00805f9b34fb
Unknown

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service001b/char001e/desc0020
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Primary Service

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0021
0000ffe0-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0022
0000ffel-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0024
0000ffe2-0000-1000-8000-00805f9b34fb
Unknown

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0021/char0024/desc0026
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Primary Service

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027
0000fee7-0000-1000-8000-00805f9b34fb
Tencent Holdings Limited

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0028
0000fec7-0000-1000-8000-00805f9b34fb
Apple, Inc.



Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a
0000fec8-0000-1000-8000-00805f9b34fb
Apple, Inc.

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a/desc002c
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002d
0000fec9-0000-1000-8000-00805f9b34fb
Apple, Inc.

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002f
0000feal-0000-1000-8000-00805f9b34fb
Intrepid Control Systems, Inc.

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002f/desc0031
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0032
0000fea2-0000-1000-8000-00805f9b34fb
Intrepid Control Systems, Inc.

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char0032/desc0034
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration

Primary Service

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035
0000cc00-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0036
0000cc02-0000-1000-8000-00805f9b34fb
Unknown

Characteristic

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0038
0000cc03-0000-1000-8000-00805f9b34fb
Unknown

Descriptor

/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0038/desc003a
00002902-0000-1000-8000-00805f9b34fb
Client Characteristic Configuration



```
Characteristic
  /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003b
  0000cc04-0000-1000-8000-00805f9b34fb
  Unknown

Descriptor
  /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003b/desc003d
  00002902-0000-1000-8000-00805f9b34fb
  Client Characteristic Configuration

Characteristic
  /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003e
  0000cc05-0000-1000-8000-00805f9b34fb
  Unknown

Descriptor
  /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char003e/desc0040
  00002902-0000-1000-8000-00805f9b34fb
  Client Characteristic Configuration

Characteristic
  /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0041
  0000cc06-0000-1000-8000-00805f9b34fb
  Unknown

[MSM3]# select-attribute /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a/desc002c
[MSM3:/service0027/char002a/desc002c]# attribute-info
Descriptor - Client Characteristic Configuration
  UUID: 00002902-0000-1000-8000-00805f9b34fb
  Characteristic: /org/bluez/hci0/dev_BA_03_64_29_12_90/service0027/char002a
[MSM3:/service0027/char002a/desc002c]# select-attribute
/org/bluez/hci0/dev_BA_03_64_29_12_90/service0035/char0038
[MSM3:/service0035/char0038]# attribute-info
Characteristic - Unknown
  UUID: 0000cc03-0000-1000-8000-00805f9b34fb
  Service: /org/bluez/hci0/dev_BA_03_64_29_12_90/service0035
  Notifying: no
  Flags: read
  Flags: notify

[DEL] Device D8:E0:E1:11:61:99 [TV] Samsung 7 Series (65)
[DEL] Device 78:5B:93:00:C5:A9 78-5B-93-00-C5-A9
[DEL] Device E7:EB:88:EB:88:5D Amazfit GTS
[DEL] Device D9:1C:99:0B:DB:D1 D9-1C-99-0B-DB-D1
[DEL] Device E6:E4:B3:9F:71:BD E6-E4-B3-9F-71-BD
[DEL] Device D7:BA:36:EB:4B:CD D7-BA-36-EB-4B-CD
[MSM3:/service0035/char0038]# quit
root@cinnamon:/home/pasg#
```



22. Existe um conjunto bastante vasto de aplicações para dispositivos móveis, que permitem o acesso aos serviços e características dos dispositivos BLE, através de dispositivos móveis e de uma forma gráfica e bastante intuitiva, como são exemplos o nRF Connect ou BLE Scanner.

